

CSCE 420 HW3

1. Translating the following sentence into First-Order logic

- Bowling balls are sporting equipment

$\forall x \text{ bowlingBall}(x) \rightarrow \text{sportingEquipment}(x)$

- All horses have a higher speed than any frog

$\forall x \text{ Horse}(x) \rightarrow (\forall y \text{ Frog}(y) \rightarrow \text{speed}(x) \wedge \text{speed}(y) \wedge \text{speedOf}(x) > \text{speedOf}(y))$

- All domesticated horses have an owner

$\forall x \text{ Horse}(x) \wedge \text{domestic}(x) \rightarrow \exists y \text{ Owner}(y)$

- The rider of a horse can be different than the owner

$\forall x \forall y \text{ Horse}(x) \wedge \text{Rider}(y) \wedge \text{Rides}(x,y) \rightarrow \exists z \text{ Owner}(z) \wedge \text{Owns}(x,z) \wedge (z \neq y)$

- A finger is any digit on a hand other than the thumb

$\forall x \text{ partOf}(x, \text{hand}) \wedge \neg \text{thumb}(x) \leftrightarrow \text{finger}(x)$

- an isosceles triangle is defined as a polygon with 3 edges connected at 3 vertices, where 2 (but not 3) edges have the same length

$\forall y \forall z \forall w \forall u \forall t \forall v \text{ hasEdgesAndVertices}(y,z,w,u,t,v) \rightarrow \text{Edge}(w) \wedge \text{Edge}(y) \wedge \text{Edge}(z) \wedge \text{Vertices}(u) \wedge \text{Vertices}(v) \wedge \text{Vertices}(t)$

$\forall y \forall z \forall w \text{ onlyTwoSameEdges}(y,z,w) \rightarrow (\text{SameLength}(y,z) \wedge \text{SameLength}(x,z) \wedge \neg \text{SameLength}(y,x)) \vee (\text{SameLength}(y,x) \wedge \text{SameLength}(x,z) \wedge \neg \text{SameLength}(y,z)) \vee (\text{SameLength}(y,x) \wedge \text{SameLength}(y,z) \wedge \neg \text{SameLength}(x,z))$

$\forall x \text{ IsoscelesTriangle}(x) \rightarrow \text{Polygon}(x) \wedge (\exists w \exists y \exists z \exists u \exists t \exists v \text{ hasEdgesAndVertices}(y,z,w,u,t,v) \wedge x = (y,w,z,u,t,v) \wedge \text{onlyTwoSameEdges}(y,z,w))$

2. Convert the following first-order logic sentence into CNF:

$$\forall x \text{ person}(x) \wedge [\exists z \text{ petOf}(x,z) \wedge \forall y \text{ petOf}(x,y) \rightarrow \text{dog}(y)] \rightarrow \text{doglover}(x)$$

$$\forall x \text{ person}(x) \wedge [\exists z \text{ petOf}(x,z) \wedge \forall y \neg \text{petOf}(x,y) \vee \text{dog}(y)] \rightarrow \text{doglover}(x)$$

$$\forall x \text{ person}(x) \wedge [\exists z \text{ petOf}(x,z) \wedge (\forall y \neg \text{petOf}(x,y) \vee \text{dog}(y))] \rightarrow \text{doglover}(x)$$

$$\forall x \neg [\text{person}(x) \wedge [\exists z \text{ petOf}(x,z) \wedge (\forall y \neg \text{petOf}(x,y) \vee \text{dog}(y))]] \vee \text{doglover}(x)$$

$$\forall x \neg \text{person}(x) \vee \neg [\exists z \text{ petOf}(x,z) \wedge (\forall y \neg \text{petOf}(x,y) \vee \text{dog}(y))] \vee \text{doglover}(x)$$

$$\forall x \neg \text{person}(x) \vee [\neg \exists z \neg \text{petOf}(x,z) \vee \neg (\forall y \neg \text{petOf}(x,y) \vee \text{dog}(y))] \vee \text{doglover}(x)$$

$$\forall x \neg \text{person}(x) \vee [\neg \exists z \neg \text{petOf}(x,z) \vee (\neg \forall y \neg \neg \text{petOf}(x,y) \wedge \neg \neg \text{dog}(y))] \vee \text{doglover}(x)$$

$$\forall x \neg \text{person}(x) \vee [\forall z \neg \neg \text{petOf}(x,z) \vee (\exists y \neg \neg \neg \text{petOf}(x,y) \wedge \neg \neg \neg \text{dog}(y))] \vee \text{doglover}(x)$$

$$\forall x \neg \text{person}(x) \vee [\forall z \text{ petOf}(x,z) \vee (\exists y \neg \text{petOf}(x,y) \wedge \text{dog}(y))] \vee \text{doglover}(x)$$

$$\forall x \neg \text{person}(x) \vee [\forall z (\text{petOf}(x,z) \vee \exists y \neg \text{petOf}(x,y)) \wedge (\exists y \text{petOf}(x,z) \vee \text{dog}(y))] \vee \text{doglover}(x)$$

$$\forall x \neg \text{person}(x) \vee [(\text{petOf}(x,F(x)) \vee \neg \text{petOf}(x,G(x))) \wedge (\text{petOf}(x,F(x)) \vee \text{dog}(G(x)))] \vee \text{doglover}(x)$$

$$\neg \text{person}(x) \vee [(\text{petOf}(x,F(x)) \vee \neg \text{petOf}(x,G(x))) \wedge (\text{petOf}(x,F(x)) \vee \text{dog}(G(x)))] \vee \text{doglover}(x)$$

$$[\neg \text{person}(x) \vee \text{doglover}(x) \text{petOf}(x,F(x)) \vee \neg \text{petOf}(x,G(x))] \wedge [\neg \text{person}(x) \vee \text{doglover}(x) \text{petOf}(x,F(x)) \vee \text{dog}(G(x))]$$

Final CNF form:

$$[\neg \text{person}(x) \vee \text{doglover}(x) \text{petOf}(x,F(x)) \vee \neg \text{petOf}(x,G(x))] \wedge [\neg \text{person}(x) \vee \text{doglover}(x) \text{petOf}(x,F(x)) \vee \text{dog}(G(x))]$$

3. Determine whether or not the following pairs of predicates are unifiable. If they are, give the most-general unifier and show the result of applying the substitution to each predicate. If they are not unifiable, indicate why. Capital letters represent variables; constants and function names are lowercase. For example, 'loves(A,hay)' and 'loves(horse,hay)' are unifiable, the unifier is $u=\{A/horse\}$, and the unified expression is 'loves(horse,hay)' for both.

• owes(owner(X),citibank,cost(X)) owes(owner(ferrari),Z,cost(Y))

These pair of predicates is unifiable.

$u = \{X/ferrari, Z/citibank, Y/ferrari\}$

Unified expression: owes(owner(ferrari),citibank,cost(ferrari))

• gives(bill, jerry, book21) gives(X,brother(X),Z)

These predicates are not unifiable. Because in the second predicate, brother(X) which contains X variable cannot be unified to a constant term "jerry". It would have been unifiable for the second term to be brother(bill).

• opened(X,result(open(X),s0))) opened(toolbox,Z)

$u = \{X/toolbox, Z/result(open(X), s0)\}$

Unified expression: opened(toolbox, result(open(toolbox),s0))

4. Consider the following situation:

a) Translate these sentences to First-Order Logic.

1. **Marcus is a Pompeian:** $\text{Pompeian}(\text{Marcus})$
2. **All Pompeians are Romans:** $\forall x \text{ Pompeian}(x) \rightarrow \text{Roman}(x)$
3. **Caesar is a ruler:** $\text{Ruler}(\text{Caesar})$
4. **All Romans are either loyal to Caesar or hate Caesar (but not both):** $\forall x \text{ Roman}(x) \rightarrow (\text{Loyal}(x, \text{Caesar}) \wedge \neg \text{Hate}(x, \text{Caesar})) \vee (\neg \text{Loyal}(x, \text{Caesar}) \wedge \text{Hate}(x, \text{Caesar}))$
5. **Everyone is loyal to someone:** $\forall x \exists y \text{ Loyal}(x, y)$
6. **People only try to assassinate rulers they are not loyal to:** $\forall x \forall y \text{ Ruler}(y) \wedge \text{Assassinate}(x, y) \rightarrow \neg \text{Loyal}(x, y)$
7. **Marcus tries to assassinate Caesar:** $\text{Assassinate}(\text{Marcus}, \text{Caesar})$

b) Prove that Marcus hates Caesar using Natural Deduction. Label all derived sentences with the ROI and which prior sentences and unifier were used.

8. $\text{Roman}(\text{Marcus})$ [MP, 1, 2] $u = \{x/\text{Marcus}\}$
9. $\text{Ruler}(\text{Caesar}) \wedge \text{Assassinate}(\text{Marcus}, \text{Caesar})$ [AndIntroduction, 3, 7]
10. $\neg \text{Loyal}(\text{Marcus}, \text{Caesar})$ [MP, 6, 9] $u = \{y/\text{Caesar}, x/\text{Marcus}\}$
11. $(\text{Loyal}(\text{Marcus}, \text{Caesar}) \wedge \neg \text{Hate}(\text{Marcus}, \text{Caesar})) \vee (\neg \text{Loyal}(\text{Marcus}, \text{Caesar}) \wedge \text{Hate}(\text{Marcus}, \text{Caesar}))$ [MP, 8, 7] $u = \{x/\text{Marcus}\}$
13. $\text{Hate}(\text{Marcus}, \text{Caesar})$ [AndElimination, 10, 11]

c) Convert all the sentences into CNF

1. $\text{Pompeian}(\text{Marcus})$
2. $\neg \text{Pompeian}(x) \vee \text{Roman}(x)$
3. $\text{Rule}(\text{Caesar})$
4. $\neg \text{Roman}(x) \vee ((\text{Loyal}(x, \text{Caesar}) \wedge \text{Hate}(x, \text{Caesar})) \vee (\neg \text{Loyal}(x, \text{Caesar}) \wedge \text{Hate}(x, \text{Caesar})))$
5. $\text{Loyal}(x, f(x))$
6. $\neg \text{Ruler}(y) \vee \neg \text{Assassinate}(x, y) \vee \neg \text{Loyal}(x, y)$
7. $\text{Assassinate}(\text{Marcus}, \text{Caesar})$
8. $\neg \text{Hate}(\text{Marcus}, \text{Caesar})$ [Negation of query]

d) Prove that Marcus hates Caesar using Resolution Refutation.

New Clauses	Annotation
9. Roman(Marcus)	Reso on 1 and 2
10. $\neg \text{Ruler}(\text{Caesar}) \vee \neg \text{Loyal}(\text{Marcus}, \text{Caesar})$	Reso on 7 and 6 {x/Marcus, y/Caesar}
11. $\neg \text{Loyal}(\text{Marcus}, \text{Caesar})$	Reso on 11 and 3
12. $((\text{Loyal}(x, \text{Caesar}) \wedge \text{Hate}(x, \text{Caesar}))$	Reso on 9 and 4
13. $\text{Hate}(x, \text{Caesar}))$	Reso on 11 and 12
14. Proof succeeds. Derived empty clause	Reso on 8 and 13

5. Write a KB in First-Order Logic with rules/axioms for...

a. **Map-coloring** – every state must be exactly 1 color, and adjacent states must be different colors. Assume possible colors are states are defined using unary predicate like color(red) or state(WA). To say a state has a color, use a binary predicate, e.g. 'color(WA,red)'.

1. state(WA)
2. state(NT)
3. state(SA)
4. state(Q)
5. state(NSW)
6. state(V)
7. state(T)
8. color(R)
9. color(G)
10. color(B)
11. neigh(WA,NT)
12. neigh(WA,SA)
13. neigh(NT,SA)
14. neigh(NT,Q)
15. neigh(SA,Q)
16. neigh(SA,NSW)
17. neigh(SA,V)
18. neigh(Q,NSW)
19. neigh(NSW,V)
20. $\forall s \text{ state}(s) \rightarrow \exists c \text{ color}(c) \wedge \text{hasColor}(s,c)$
21. $\forall s \forall c \forall d \text{ state}(s) \wedge \text{hasColor}(s,c) \wedge \text{hasColor}(s,d) \rightarrow c=d$
22. $\forall s \forall t \forall c \text{ state}(s) \wedge \text{state}(t) \wedge \text{neigh}(s,t) \wedge \text{hasColor}(s,c) \rightarrow \neg \text{hasColor}(t,c)$

B. **Sammy's Sport Shop** – include implications of facts like obs(1,W) or label(2,B), as well as constraints about the boxes and colors. Use predicate 'cont(x,q)' to represent that box x contains tennis balls of color q (where q could be W, Y, or B)

1. Color(Y)
2. Color(B)

3. Color(W)
4. Box(1)
5. Box(2)
6. Box(3)
7. $\forall x \text{ Box}(x) \rightarrow \exists c \text{ color}(c) \wedge \text{cont}(x,c)$ // each box contains atleast one color
8. $\forall s \forall c \forall d \text{ Box}(x) \wedge \text{cont}(x,c) \wedge \text{cont}(x,d) \rightarrow c=d$ each box contains at most one color
9. $\forall x \text{ Box}(x) \wedge \text{cont}(x,B) \rightarrow \text{obs}(x,W) \vee \text{obs}(x,Y)$ // a box that contains both will have an observation of white or yellow
10. $\forall b \forall q \text{ Box}(b) \wedge \text{cont}(b,q) \rightarrow \neg \text{label}(b,q)$ // a box will contain an incorrect label always
11. $\forall b \forall q \text{ Box}(b) \wedge \text{cont}(b,q) \rightarrow \text{obs}(b,q)$ // a box will have same observation as it is contain

Any optional facts can be added like obs(1,w) or label(1,b)

C. Wumpus World - (hint start by defining a helper concept 'adjacent(x,y,p,q)' which defines when a room at coordinates (x,y) is adjacent to another room at (p,q). Don't forget rules for 'stench', 'breezy', and 'safe'.

1. $\forall x \forall y \forall p \forall q \text{ adjacent}(x,y,p,q) \rightarrow ((p = \text{plus}(x,1)) \wedge (q=y) \vee ((p = \text{subtract}(x,1)) \wedge (q=y)) \vee ((p = x) \wedge (q=\text{plus}(x,1))) \vee ((p = x) \wedge (q=\text{subtract}(x,1))))$
2. $\forall p \forall q \exists x \exists y \text{ adjacent}(x,y,p,q) \wedge \text{Pit}(x,y) \rightarrow \text{Breeze}(p,q)$ // there exist some squares with a pit where all its adjacent squares have a breeze
3. $\forall p \forall q \exists x \exists y \text{ adjacent}(x,y,p,q) \wedge \text{Wumpus}(x,y) \rightarrow \text{Stench}(p,q)$ // there exists some squares with wumpus where all adjacent squares have a stench
4. $\forall p \forall q \text{ Gold}(p,q) \rightarrow \text{Glitter}(p,q)$
5. $\forall p \forall q \text{ Wumpus}(p,q) \wedge \text{Pit}(p,q) \rightarrow \neg \text{safe}(p,q)$

D. 4-Queens – assume row(1)...row(4) and col(1)...col(4) are facts; write rules that describe configurations of 4 queens such that none can attack each other, using 'queen(r,c)' to represent that there is a queen in row r and col c.

1. row(1)
2. row(2)
3. row(3)
4. row(4)
5. col(1)
6. col(2)
7. col(3)
8. col(4)
9. $\forall r \text{ row}(r) \rightarrow \exists c \text{ queen}(r,c)$ // every row has atleast 1 queen in it
10. $\forall c \text{ col}(c) \rightarrow \exists r \text{ queen}(r,c)$ // every col has atleast 1 queen in it
11. $\forall r \forall c \forall d \text{ row}(r) \wedge \text{col}(c) \wedge \text{col}(d) \wedge \text{queen}(r,c) \wedge \text{queen}(r,d) \rightarrow c=d$ // every row has at most one queen

12. $\forall r \forall c \forall d \text{ row}(r) \wedge \text{col}(c) \wedge \text{row}(d) \wedge \text{queen}(r,c) \wedge \text{queen}(d,c) \rightarrow r=d$ // every col has at most one queen
13. $\forall r \forall c \text{ queen}(r,c) \rightarrow \exists a \exists b \neg \text{queen}(a,b) \wedge (a = \text{Plus}(r,i)) \wedge (i \neq 0) \wedge (b = \text{Plus}(r,i)) \wedge \text{row}(a) \wedge \text{row}(b)$ // if there is a queen on r,c that means there is no queen in the top right diagonal
14. $\forall r \forall c \text{ queen}(r,c) \rightarrow \exists a \exists b \neg \text{queen}(a,b) \wedge (a = \text{Plus}(r,i)) \wedge (i \neq 0) \wedge (b = \text{subtract}(c,i)) \wedge \text{row}(a) \wedge \text{row}(b)$ // if there is a queen on r,c that means there is no queen in the top left diagonal
15. $\forall r \forall c \text{ queen}(r,c) \rightarrow \exists a \exists b \neg \text{queen}(a,b) \wedge (a = \text{subtract}(r,i)) \wedge (i \neq 0) \wedge (b = \text{Plus}(c,i)) \wedge \text{row}(a) \wedge \text{row}(b)$ // if there is a queen on r,c that means there is no queen in the bottom right diagonal
16. $\forall r \forall c \text{ queen}(r,c) \rightarrow \exists a \exists b \neg \text{queen}(a,b) \wedge (a = \text{subtract}(r,i)) \wedge (i \neq 0) \wedge (b = \text{subtract}(c,i)) \wedge \text{row}(a) \wedge \text{row}(b)$ // if there is a queen on r,c that means there is no queen in the bottom left diagonal